

Phase Equilibria Studies of Cu-O-Si Systems in Equilibrium with Air and Metallic Copper and Cu-Me-O-Si Systems (Me = Ca, Mg, Al, and Fe) in Equilibrium with Metallic Copper

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The pseudo-binary phase diagrams in the Cu-O-Si system in equilibrium with air and metallic copper have been constructed. Equilibration at high temperature in an appropriate containment material, followed by rapid quenching and measurement of phase compositions using an electron probe x-ray microanalysis (EPMA) technique, was carried out to obtain the phase equilibria data. The investigation has been extended to characterize the effects of CaO, MgO, Al₂O₃, and “Fe₂O₃” on the liquidus isotherms of tridymite in equilibrium with metallic copper at temperatures of 1473 K and 1573 K (1200 °C and 1300 °C).

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I. INTRODUCTION

PREVIOUS studies^[1–5] of the Cu-O-Si system in equilibrium with air and copper metal have shown that in the high-copper region, the phase equilibria in each case can be represented by a pseudo-binary phase diagram having a simple eutectic reaction with two primary phase fields of cuprite (Cu₂O) and tridymite (SiO₂). No binary compound is formed in the system.

The phase stability in the Cu-O system has been assessed previously.^[6–9] Above 1300 K (1027 °C), Cu₂O is the stable solid phase in air rather than CuO. The melting temperature of Cu₂O in equilibrium with air has been reported to be approximately 1397 K (1124 °C). The phase equilibria in the “Cu₂O”-SiO₂ system in equilibrium with air have been investigated previously by an equilibration-quenching technique, in conjunction with optical microscopy, X-ray diffraction analysis, and wet chemical analysis,^[1,4,5] and by thermo-gravimetric^[2] techniques. Berezhnoi *et al.*^[1] reported eutectic temperature at 1333 K (1060 °C), which agrees well with that reported by Gadalla *et al.*^[2] Landolt^[4] and Peddada and Gaskell^[5] reported the liquid compositions at tridymite (SiO₂) saturation at 1473 K and 1573 K (1200 °C and 1300 °C, respectively). There are, however, significant differences^[1,4,5] in the reported liquidus of the tridymite primary phase field.

The melting temperature of “Cu₂O” in equilibrium with metallic copper has been reported^[6–8] to be 1496 ± 2 K (1223 °C). The liquidus values of tridymite in equilibrium with metallic copper reported by Kuxmann and Kurre^[3] and Landolt^[4] from equilibration-quenching experiments in a silica crucible under

inert N₂ gas atmosphere and an evacuated silica tube sealed under vacuum, respectively, are in agreement. The eutectic point was reported by Kuxmann and Kurre^[3] at a temperature of 1459 K (1186 °C) at approximately 89.5 wt pct Cu₂O and 10.5 wt pct SiO₂.

In the present study, the phase equilibria of the Cu-O-Si system in equilibrium with air have been investigated to address differences between results reported by previous researchers. The phase equilibria of the Cu-O-Si system in equilibrium with metallic copper have been revisited to confirm the positions of tridymite liquidus and eutectic point and have been extended to study the effects of CaO, MgO, Al₂O₃, and “Fe₂O₃” on the tridymite liquidus.

II. EXPERIMENTAL TECHNIQUE AND PROCEDURE

A. Preparation of Initial Mixtures

The initial mixtures were prepared from high-purity (99.9+ wt pct purity) oxide powders of SiO₂, Cu₂O, MgO, Al₂O₃, Fe₂O₃, CuO (99.3+ wt pct purity), CaO (calcined from CaCO₃ with 99.0 wt pct purity), and Cu (99.7 wt pct purity) (supplied by Alfa Aesar, MA) by mixing thoroughly in an agate mortar pestle. The initial mixture bulk compositions were selected to obtain liquid slag in equilibrium with one or more solid phases. The powder mixtures were pelletized and less than 0.3 g of sample was used in each equilibration experiment.

B. Selection of Containment Materials

A 10-mm × 12-mm envelope made from 0.025-mm-thick platinum foil (Figure 1(a)) was used for experiments in air. It was found that in the conditions under investigation, 10 to 19 wt pct Cu was dissolved in the platinum; although this changed the bulk composition of the mixture, the compositions of liquid and solid phases at equilibrium did not change. A quartz (SiO₂)

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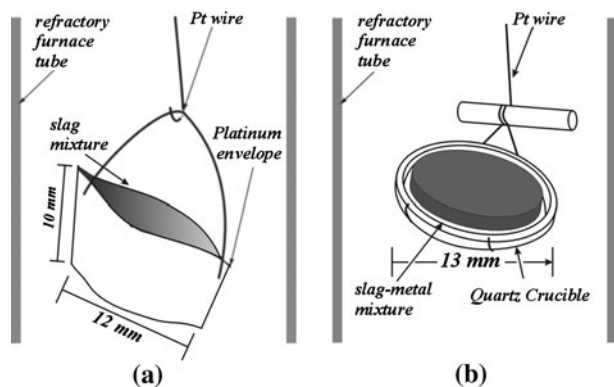


Fig. 1—Crucible designs and suspension: (a) platinum crucible and (b) quartz crucible.

crucible with a diameter of 13 mm and depth of 5 mm, made by fusing high-purity silica rod, was used for experiments in equilibrium with molten copper (see Figure 1(b)).

C. Equilibration Technique

Equilibration experiments were carried out in a vertical reaction tube (impervious recrystallized alumina, 30-mm inner diameter) within electrical resistance silicon carbide (SiC) heated furnaces (Figure 2). A working thermocouple in a recrystallized alumina sheath was placed immediately next to the sample to monitor the actual temperature of the sample. The working thermocouple was periodically calibrated against a standard thermocouple (supplied by the National Measurement Institute of Australia, NSW, Australia). The overall absolute temperature accuracy of the experiment was estimated to be within 5 K (5 °C).

The sample, hooked to the platinum wire (0.5-mm diameter), was introduced from the bottom of the reaction tube. It was kept in the cold zone, while flushing the reaction tube for more than 5 minutes with the required gas, then raised and suspended in the hot zone with 30 minutes of premelting at 25 K (25 °C) above the equilibration temperature, and finally held for equilibration at the final target temperature for the required equilibration period.

Experiments for the Cu-O-Si system in air were undertaken with the bottom and the top of the reaction tube left open to the atmosphere. For experiments in equilibrium with metallic copper, an inert high-purity argon gas (99.999 pct purity; supplied by Coregas, NSW, Australia) was continuously flushed through the gas-tight system. The argon gas may contain O_2 , as well as other impurities including N_2 , CO_2 , H_2O , CH_4 , etc., in concentrations of less than 5 ppm. The effects of these minor components in the gas on the oxidation/reduction reactions of condensed phases, or in this case on the slag compositions, were not noticeable and were not detected by EPMA analysis (electron probe X-ray microanalyzer). The final slag compositions were majorly dependent on the bulk composition of the initial mixture. The experimental conditions therefore can be considered as a

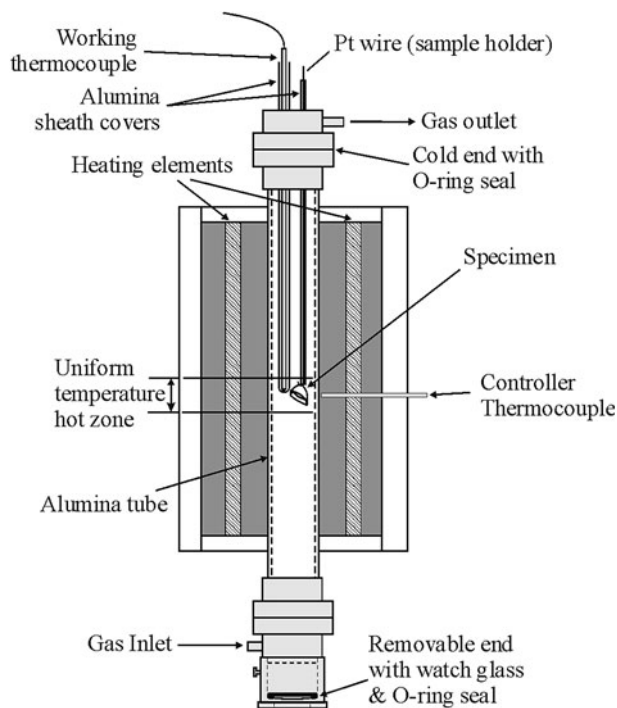


Fig. 2—Furnace design for equilibrium experiments.

closed-system condition with the effective oxygen partial pressure in the sample controlled by the equilibrium between the condensed phases, *i.e.*, slag and Cu-Fe alloy.

After the equilibration was completed, the bottom end of the reaction tube was submerged in water and opened; the specimen was then rapidly quenched into the cold NaCl-saturated water, washed thoroughly, dried on a hot plate, and mounted in epoxy resin. A polished cross section of the mounted specimen was prepared using conventional metallographic grinding and polishing techniques.

D. Electron Probe X-ray Microanalysis (EPMA)

The polished specimens were carbon-coated and their microstructures were examined by Scanning Electron Microscopy (SEM). The compositions of the phases were measured by EPMA (an electron probe X-ray microanalyzer JEOL 8200L with wavelength dispersive detectors; Japan Electron Optics Ltd., Tokyo, Japan) operated with 15-kV accelerating voltage and a 15 nA probe. The standard Duncumb-Philibert ZAF correction supplied with the probe was applied. $CaSiO_3$, MgO , Al_2O_3 , Fe_2O_3 standards (supplied by Charles M. Taylor Co., Stanford, CA), and Cu_2O standard (supplied by Structure Probe Inc. (SPI) Supplies, West Chester, PA) were used. Only the proportions of metal cations in the sample were measured by EPMA; all copper oxide and iron oxide concentrations were reported as Cu_2O and Fe_2O_3 , respectively, for presentation purposes.

Examples of typical microstructures of the quenched samples in Cu-O-Si system (backscattered scanning electron micrographs) are given in Figure 3. In the case

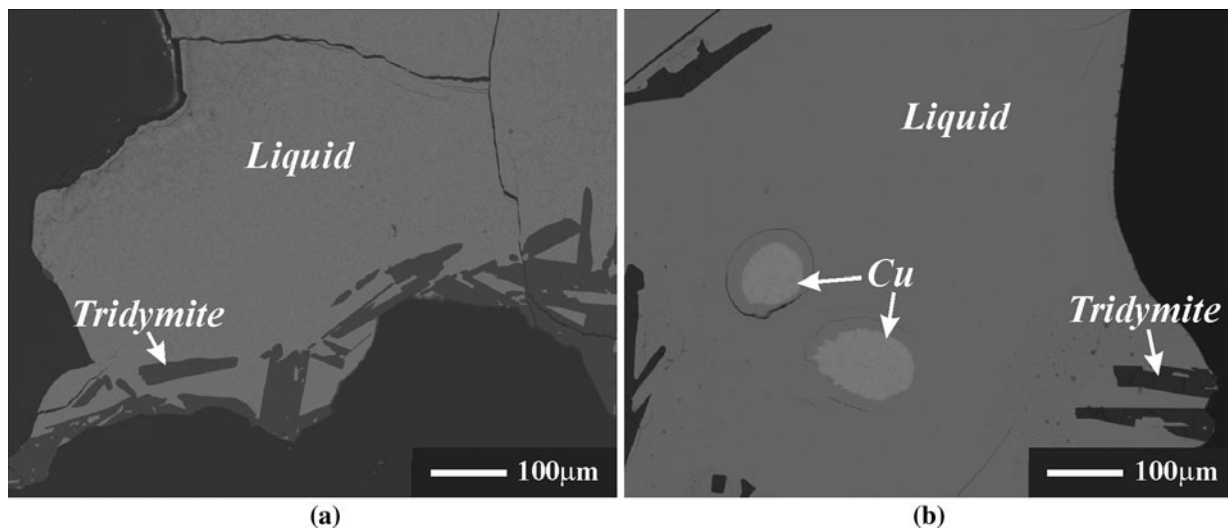


Fig. 3—Backscattered scanning electron micrographs of quenched Cu-O-Si slags saturated with tridymite (SiO_2) crystal: (a) in equilibrium with air and (b) in equilibrium with metallic copper.

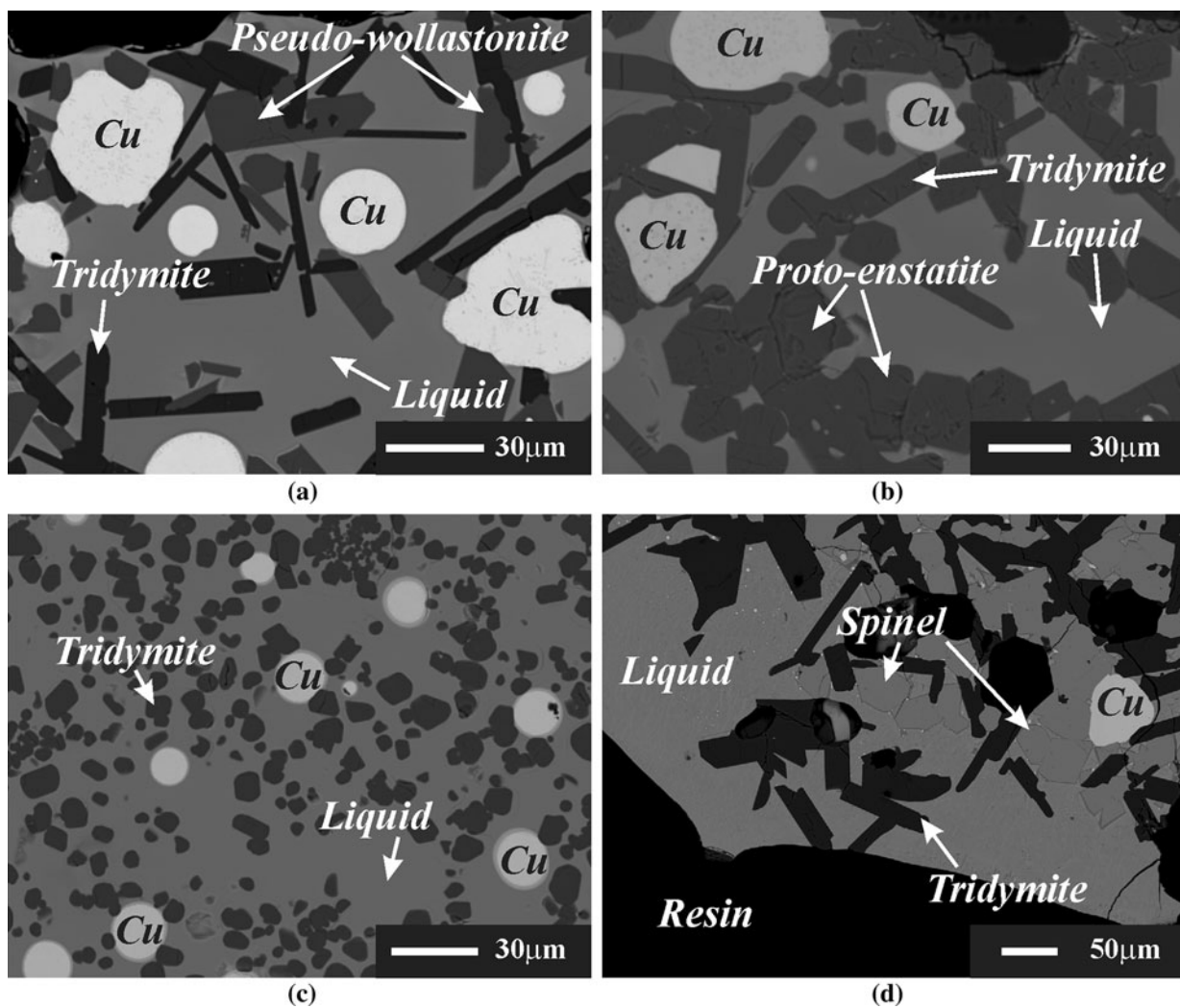


Fig. 4—Backscattered scanning electron micrographs of quenched Cu-Me-O-Si slags in equilibrium with metallic copper: (a) " Cu_2O "-CaO- SiO_2 system at tridymite (SiO_2) and pseudo-wollastonite (CaSiO_3) saturation; (b) " Cu_2O "-MgO- SiO_2 system at tridymite (SiO_2) and proto-enstatite (MgSiO_3) saturation; (c) " Cu_2O "- Al_2O_3 - SiO_2 system at tridymite (SiO_2) saturation; and (d) " Cu_2O "- Fe_2O_3 - SiO_2 system at tridymite (SiO_2) and spinel (" Fe_3O_4 ") saturation.

of high-silica liquid, a glassy phase was readily formed on quenching. In the case of low-silica liquid, the glassy homogeneous area was limited only to locations at the surface directly contacting the quenching medium. The criteria for performing an accurate, repeatable, and objective measurement by the EPMA-line analysis method described by Nikolic *et al.*^[10] were applied to obtain the average compositions of the liquid phase in the Cu-O-Si systems. There was no difficulty in the quenching of the liquid in the Cu-Me-O-Si systems; the compositions of the liquid and the crystal phases there were measured by the EPMA-point analysis method. Examples of typical microstructures of the quenched samples in Cu-Me-O-Si systems (backscattered scanning electron micrographs) are given in Figure 4.

E. Confirmation of Achievement of Equilibrium

Particular attention was paid to insure the achievement of equilibrium by (1) approaching the final equilibrium compositions from different directions, *i.e.*, by starting with two different initial oxide mixtures—one of CuO-SiO₂ and the other Cu₂O-SiO₂, (2) variation of equilibration time, (3) assessment of the compositional homogeneity of the phases by EPMA, and (4) consideration of the specific reactions that may hinder the achievement of equilibrium. Several dedicated sets of experiments were carried out for these purposes and some of the results are reported in Tables I, II, and III.

The introduction of copper oxide in the form of Cu₂O or CuO in the Cu-O-Si systems did not give appreciable differences in the final phase compositions at both conditions. It was found that the equilibrium in the Cu-O-Si systems in air at 1343 K (1070 °C) was readily achieved within 12 hours and in contact with metallic

copper at 1473 K (1200 °C), it was achieved within 6 hours.

The attainment of equilibrium in the Cu-Me-O-Si systems (Me = Ca, Mg, Al, and Fe) in equilibrium with metallic copper tested with initial mixtures prepared from CuO was achieved within 6 hours at relatively low temperatures between 1373 K and 1473 K (between 1100 °C and 1200 °C). Equilibrium at higher temperatures was achieved within a shorter period; therefore, in some cases in the Cu-Fe-O-Si system,^[11] the equilibration period of 4 hours was selected for experiments above 1473 K (1200 °C).

III. RESULTS AND DISCUSSION

A. Cu-O-Si System in Equilibrium with Air

The results of the phase equilibria experiments of the Cu-O-Si system in equilibrium with air in the tridymite (SiO₂) primary phase field at temperatures between 1343 K and 1623 K (1070 °C and 1350 °C) are reported in Table I and presented in Figure 5. The SiO₂ concentration in the liquid at the tridymite liquidus obtained in the present study is found to be higher than that reported by Berezhnoi *et al.*^[1] and Peddada and Gaskell.^[5] At 1473 K (1200 °C), the tridymite liquidus obtained by the present study, within experimental uncertainty, is in agreement with the tridymite liquidus reported by Landolt.^[4] The tridymite crystal has been found to contain significant concentrations of Cu₂O, between 1.2 and 1.6 wt pct, in solid solution (see Table I).

Cuprite and tridymite solids were observed at temperature of 1327 K (1054 °C). From the present study, it is expected that the eutectic temperature is located between 1327 K and 1343 K (1054 °C and 1070 °C).

Table I. Experimentally Determined Phase Compositions for the “Cu₂O”-SiO₂ System in Equilibrium with Air

Temperature [K(°C)]	Equilibration Time (h)	Phase	SiO ₂ (wt pct)	“Cu ₂ O” (wt pct)
1327 (1054)	6	cuprite	0	100.0
		tridymite	98.5	1.5
1343 (1070)	12	liquid	12.2	87.8
		tridymite	98.5	1.5
1343 (1070)	24	liquid	12.6	87.4
		tridymite	98.5	1.5
1373 (1100)	12	liquid	12.7	87.3
		tridymite	98.4	1.6
1423 (1150)	12	liquid	14.0	86.0
		tridymite	98.6	1.4
1473 (1200)	12	liquid	15.2	84.8
		tridymite	98.7	1.3
1523 (1250)	12	liquid	16.6	83.4
		tridymite	98.7	1.3
1523 (1250)	12	liquid	17.4	82.6
		tridymite	98.8	1.2
1573 (1300)	12	liquid	17.5	82.5
		tridymite	98.8	1.2
1573 (1350)	12	liquid	20.3	79.7
		tridymite	98.8	1.2

Table II. Experimentally Determined Phase Compositions for the “Cu₂O”-SiO₂ System in Equilibrium with Metallic Copper

Temperature [K(°C)]	Equilibration Time (h)	Phase	SiO ₂ (wt pct)	“Cu ₂ O” (wt pct)
1461 (1188)	12	cuprite	0	100.0
		tridymite	98.5	1.5
1473 (1200)	6	liquid	12.0	88.0
		tridymite	98.7	1.3
1473 (1200)	6	liquid	11.7	88.3
		tridymite	98.7	1.3
1473 (1200)	24	liquid	12.0	88.0
		tridymite	98.7	1.3
1523 (1250)	6	liquid	12.7	87.3
		tridymite	98.8	1.2
1523 (1250)	6	liquid	12.6	87.4
		tridymite	99.0	1.0
1573 (1300)	6	liquid	13.9	86.1
		tridymite	98.9	1.1
1623 (1350)	12	liquid	15.4	84.6
		tridymite	99.0	1.0
1673 (1400)	6	liquid	16.9	83.1
		tridymite	98.9	1.1
1673 (1400)	6	liquid	17.9	82.1
		tridymite	99.1	0.9

Table III. Experimentally Determined Phase Compositions for the “Cu₂O”-SiO₂-MeO_x Systems (MeO_x = CaO, MgO, and Al₂O₃) in Equilibrium with Metallic Copper

Temperature [K(°C)] CaO Addition	Equilibration Time (h)	Phase	Composition (wt pct)		
			CaO	SiO ₂	“Cu ₂ O”
1473 (1200)	6	liquid	3.9	24.1	72.0
		tridymite	0.0	98.7	1.3
1473 (1200)	6	liquid	7.7	31.8	60.5
		tridymite	0.0	98.8	1.2
1473 (1200)	6	liquid	11.5	37.8	50.7
		tridymite	0.0	98.9	1.1
1473 (1200)	6	liquid	16.5	45.0	38.5
		tridymite	0.1	99.1	0.8
		pseudo-wollastonite	47.8	51.2	1.0
1473 (1200)	6	liquid	16.7	45.2	38.1
		tridymite	0.1	99.1	0.8
		pseudo-wollastonite	47.6	51.5	0.9
1473 (1200)	12	liquid	16.9	45.0	38.1
		tridymite	0.1	99.0	0.9
		pseudo-wollastonite	48.1	50.8	1.1
1573 (1300)	24	liquid	7.2	34.1	58.7
		tridymite	0.0	98.8	1.2
1573 (1300)	6	liquid	10.4	41.3	48.3
		tridymite	0.0	98.8	1.2
1573 (1300)	6	liquid	17.3	49.1	33.6
		tridymite	0.0	98.9	1.1
1573 (1300)	6	liquid	24.3	53.7	22.0
		tridymite	0.1	99.2	0.7
		pseudo-wollastonite	47.9	51.3	0.8
MgO AdditionM			MgO	SiO ₂	“Cu ₂ O”
1473 (1200)	6	liquid	1.9	18.0	80.1
		tridymite	0.0	98.7	1.3
		proto-enstatite	39.2	57.6	3.2
1473 (1200)	12	liquid	1.7	17.8	80.5
		tridymite	0.0	98.6	1.4
		proto-enstatite	38.6	58.5	2.9
1473 (1200)	12	liquid	1.6	18.0	80.4
		tridymite	0.0	98.7	1.3
		proto-enstatite	38.5	58.3	3.2
1573 (1300)	6	liquid	4.3	28.2	67.5
		tridymite	0.0	98.8	1.2
1573 (1300)	6	liquid	8.4	36.7	54.9
		tridymite	0.0	98.9	1.1
1573 (1300)	6	liquid	9.5	41.6	48.9
		tridymite	0.0	98.8	1.2
		proto-enstatite	39.0	58.7	2.3
1573 (1300)	6	liquid	9.5	42.7	47.8
		tridymite	0.0	98.8	1.2
		proto-enstatite	38.8	59.0	2.2
Al ₂ O ₃ Addition			Al ₂ O ₃	SiO ₂	“Cu ₂ O”
1473 (1200)	6	liquid	2.9	20.8	76.3
		tridymite	0.5	97.4	2.1
1473 (1200)	6	liquid	5.2	27.4	67.4
		tridymite	0.7	96.9	2.4
1473 (1200)	6	liquid	8.1	37.7	54.2
		tridymite	0.8	96.8	2.4
1473 (1200)	6	liquid	15.6	64.9	19.5
		tridymite*	—	—	—
1573 (1300)	6	liquid	3.8	27.6	68.6
		tridymite	0.5	97.6	1.9
1573 (1300)	6	liquid	5.2	32.7	62.1

Table III. Continued

Al ₂ O ₃ Addition			Al ₂ O ₃	SiO ₂	"Cu ₂ O"
1573 (1300)	6	tridymite	0.6	97.2	2.2
		liquid	8.4	46.0	45.6
1573 (1300)	6	tridymite	0.8	96.8	2.4
		liquid	12.3	65.3	22.4
1573 (1300)	6	tridymite*	—	—	—
		liquid	14.6	68.4	17.0
		tridymite*	—	—	—

*The size of tridymite crystals was too small for accurate analysis.

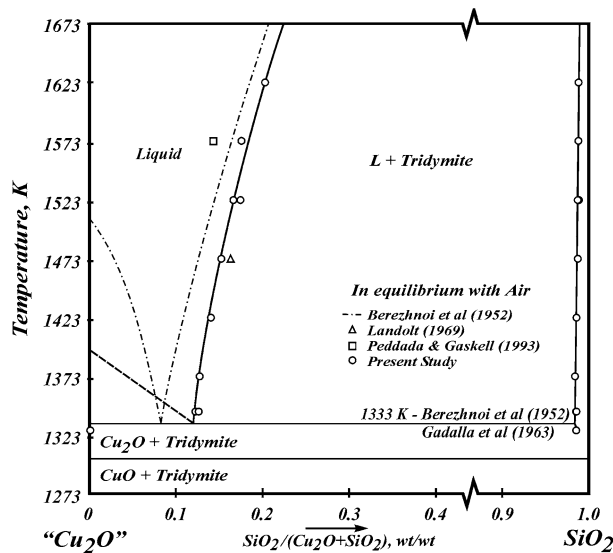


Fig. 5—Pseudo-binary phase diagram of the "Cu₂O"-SiO₂ system in equilibrium with air ($p_{O_2} = 0.21$ atm).

This supports the findings of the eutectic temperature reported by Bereznoi *et al.*^[1] and Gadalla *et al.*^[2] at 1333 K (1060 °C). The Cu₂O melting point in air has been selected at 1397 K (1124 °C), which was based on the assessment of the Cu-O system.^[6-9] The compositions of the liquids in the cuprite (Cu₂O) primary phase field were not determined in the present study.

B. Cu-O-Si System in Equilibrium with Metallic Copper

The phase equilibria of the Cu-O-Si system in equilibrium with metallic copper in the tridymite primary phase field have been investigated at temperatures between 1473 K and 1673 K (1200 °C and 1400 °C). The results of the EPMA measurements of the system are provided in Table II and presented in a pseudo-binary diagram of "Cu₂O"-SiO₂ in Figure 6. The tridymite liquidus data obtained in the present study are in general agreement with the tridymite liquidus determined by previous investigators.^[3,4] The maximum solubility of copper oxide in the tridymite phase varies between 0.9 and 1.5 wt pct "Cu₂O." The melting point of Cu₂O in equilibrium with metallic copper of 1496 K $\pm$ 2 K (1223 °C) shown in Figure 6 is

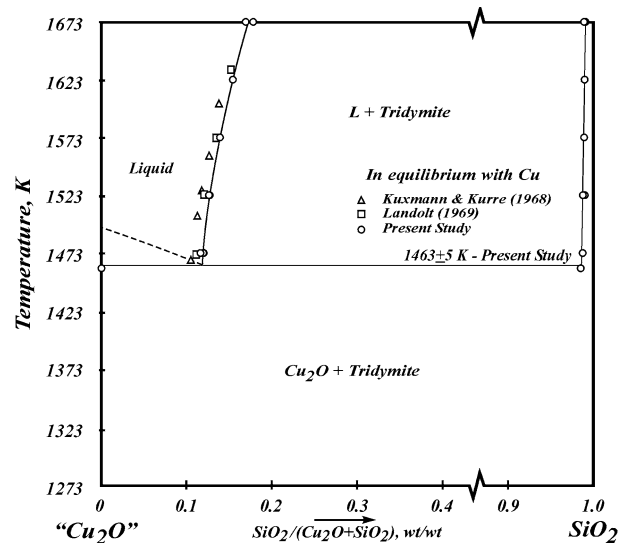


Fig. 6—Pseudo-binary phase diagram of the "Cu₂O"-SiO₂ system in equilibrium with metallic copper.

based on the thermodynamic evaluation of the Cu-O system.^[6-8]

A special set of experiments was performed to determine the eutectic point of the system in the presence of copper. The first experiment was carried out by equilibrating a Cu-O-Si mixture with excess copper in a silica crucible at 1465 K (1192 °C) for 12 hours followed by quenching. Cuprite, tridymite, and copper metal were found in this sample. Formation of liquid slag was also observed in the sample, which indicates that the equilibration temperature was above the eutectic point. The second experiment was carried out by pre-equilibrating the Cu-O-Si mixture with excess copper in silica crucible at 1473 K (1200 °C) for 6 hours, followed by equilibration at 1461 K (1188 °C) for 12 hours, and then quenching. This sample contained predominantly liquid slag, copper metal, and tridymite crystal. A small fraction of cuprite was detected in the sample; this possibly formed during the slow cooling of slag from 1473 K to 1461 K (1200 °C to 1188 °C). The results of the experiments indicate that the eutectic temperature of the system is located between 1461 K and 1465 K (1188 °C and 1192 °C), or approximately at 1463 K $\pm$ 5 K (1190 °C), which is higher by

4 K than that reported by Kuxmann and Kurre.^[3] The liquid at the eutectic contains approximately 11 wt pct SiO₂.

By comparing Figures 5 and 6, it can be observed that changing from oxidizing conditions (in equilibrium with air) to reducing conditions (in equilibrium with metallic copper) leads to a reduction of the stability area of the fully liquid slag. For example, the melting temperature of copper oxide at metallic copper saturation (*i.e.*, 1496 K (1223 °C)) is greater than that in air (*i.e.*, 1397 K (1124 °C)). The eutectic temperature of the system in the presence of metallic copper is higher by 130 K than that in air. At 1573 K (1300 °C), the maximum solubility of silica in the liquid in air is 17.5 wt pct SiO₂, while with metallic copper, this limiting value is 13.9 wt pct SiO₂.

C. The Effects of CaO, MgO, Al₂O₃, and “Fe₂O₃” on Tridymite-Saturated Liquids Containing Cu-O-Si in Equilibrium with Metallic Copper

The effects of metal oxide additions on the liquidus isotherms of tridymite-saturated Cu-O-Si slags in the presence of metallic copper have been experimentally determined for 1473 K and 1573 K (1200 °C and 1300 °C). The results of the EPMA measurements for the Cu-Me-O-Si systems (where Me = Ca, Mg, and Al) are provided in Table III. Data on the Cu-Fe-O-Si system are taken from other study by the authors.^[11] The results are summarized in pseudo-ternary diagrams of “Cu₂O”-SiO₂-CaO (Figure 7), “Cu₂O”-SiO₂-MgO (Figure 8), “Cu₂O”-SiO₂-Al₂O₃ (Figure 9), and “Cu₂O”-SiO₂-“Fe₂O₃” (Figure 10). All of the binary eutectic, peritectic, and monotectic points taken from the current study (“Cu₂O”-SiO₂ binary), the literature^[10] (“Cu₂O”-CaO and “Cu₂O”-“Fe₂O₃”), and FactSage computer software^[12] are included in the figures.

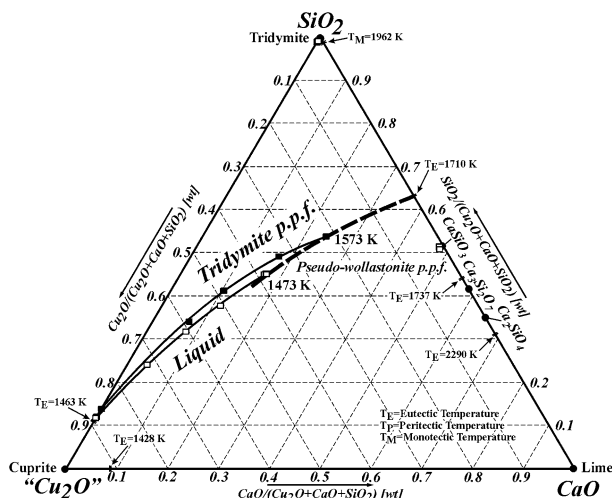


Fig. 7—Tridymite liquidus isotherms in the “Cu₂O”-SiO₂-CaO system in equilibrium with metallic copper at 1473 K and 1573 K (1200 °C and 1300 °C). The dashed line indicates the boundary between the tridymite (SiO₂) and the pseudo-wollastonite (CaSiO₃) primary phase fields.

The extent of the tridymite primary phase field in the “Cu₂O”-SiO₂-CaO system, shown in Figure 7, varies significantly from low-silica concentrations in the “Cu₂O”-SiO₂ binary (bounded by the cuprite (Cu₂O) primary phase field) to high silica in the CaO-SiO₂ binary (bounded by the pseudo-wollastonite (CaSiO₃) primary phase field). The lower limits of “Cu₂O” solubility in tridymite-saturated liquids at 1473 K and 1573 K (1200 °C and 1300 °C) coincide with the formation of the pseudo-wollastonite (CaSiO₃) phase. The estimated position of the tridymite (SiO₂)–pseudo-wollastonite (CaSiO₃) eutectic from the CaO-SiO₂ binary^[12] to the “Cu₂O”-SiO₂-CaO ternary is marked on the figure. In the tridymite primary phase field, the addition of CaO to “Cu₂O”-SiO₂ slags results in the dramatic decrease in liquidus temperature; the effect is that the

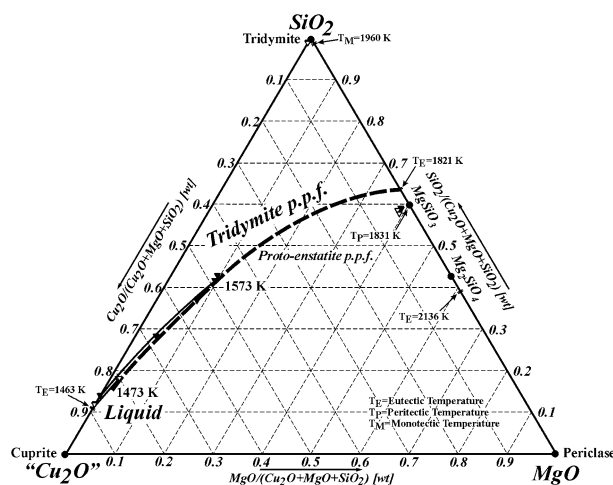


Fig. 8—Tridymite liquidus isotherms in the “Cu₂O”-SiO₂-MgO system in equilibrium with metallic copper at 1473 K and 1573 K (1200 °C and 1300 °C). The dashed line indicates the boundary between the tridymite (SiO₂) and the proto-enstatite (MgSiO₃) primary phase fields.

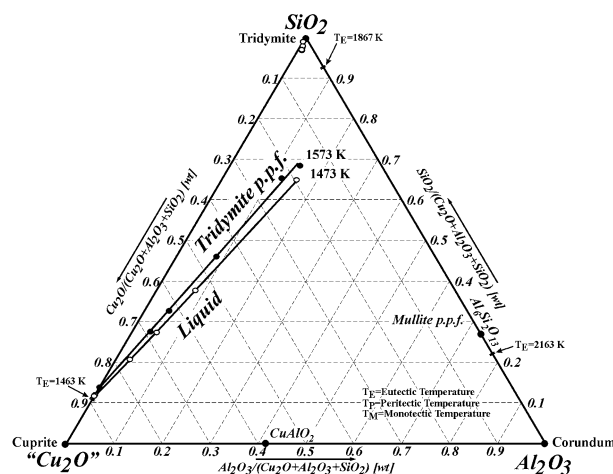


Fig. 9—Tridymite liquidus isotherms in the “Cu₂O”-SiO₂-Al₂O₃ system in equilibrium with metallic copper at 1473 K and 1573 K (1200 °C and 1300 °C).

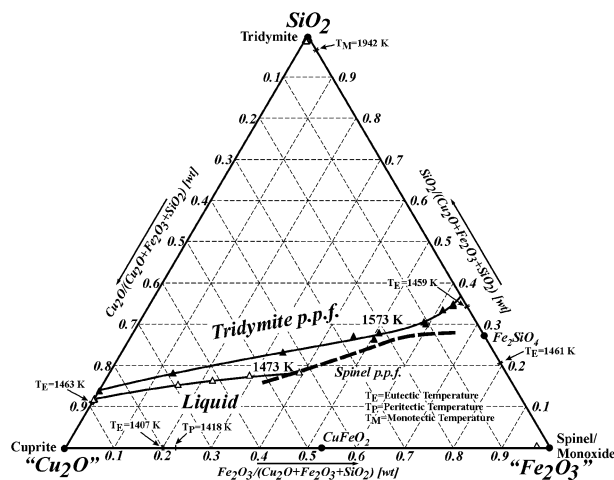


Fig. 10—Tridymite liquidus isotherms in the “Cu₂O”-SiO₂–“Fe₂O₃” system in equilibrium with metallic copper-iron alloy at 1473 K and 1573 K (1200 °C and 1300 °C). The dashed line indicates the boundary between the tridymite (SiO₂) and the spinel (“Fe₃O₄”) primary phase fields.^[1]

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Figure 8 shows the liquidus measurements obtained for the “Cu₂O”-SiO₂-MgO system at copper metal saturation. The trends observed in this system are similar to those observed in the CaO-containing system. The lower limits of “Cu₂O” solubility in tridymite-saturated liquids at 1473 K and 1573 K (1200 °C and 1300 °C) are determined by the formation of the protoenstatite (MgSiO₃) phase. The estimated position of the tridymite (SiO₂)-protoenstatite (MgSiO₃) eutectic from the MgO-SiO₂ binary^[12] to the “Cu₂O”-SiO₂-MgO ternary is marked on the figure.

The influence of the trivalent metal oxide of Al_2O_3 on the composition of tridymite liquidus in equilibrium with metallic copper is provided in Figure 9. At 1473 K and 1573 K (1200 °C and 1300 °C), the tridymite primary phase extends from the “ Cu_2O ”- SiO_2 binary to close to the Al_2O_3 - SiO_2 join. The lower limits of copper oxide solubility at these temperatures have not yet been determined; this may possibly coincide with the formation of mullite ($\text{Al}_6\text{Si}_5\text{O}_{13}$).

Figure 10 shows the influence of iron oxide on the liquidus in the “Cu₂O”-SiO₂-“Fe₂O₃” slag system in equilibrium with tridymite crystal and metallic copper-iron alloy. This diagram represents a wide range of oxygen partial pressures from approximately 10^{-3.9} atm at pure metallic copper saturation (at the “Cu₂O”-SiO₂ join^[12]) to 10^{-11.8} atm at pure metallic iron saturation (at the “FeO”-SiO₂ join^[12]). As explained earlier in Section II-D, the iron oxide concentrations in the slag phase have been reported as Fe₂O₃ for presentation purposes only. Over this wide range of gas conditions, the Cu-Fe-O-Si liquid actually contains both ferric (Fe³⁺) and ferrous (Fe²⁺) ions. The ferric to ferrous ratio increases as the composition of the liquid slag moves to higher “Cu₂O” concentrations.

Previous experimental studies on the Fe-O-Si system^[13,14] have shown that spinel (“Fe₃O₄”) and

monoxide (“FeO”) primary phase fields are present in the low-silica region; similar behavior is expected in the Cu-Fe-O-Si slag system. The Cu-Fe-O-Si slag in the low-silica region has been found to have spinel (“Fe₃O₄”) as the primary phase crystal, and the boundary between tridymite and spinel primary phase fields obtained by the authors^[11] is indicated in the figure. And, as “Cu₂O” in the slag is decreased (pO₂ is decreased), the primary phase crystal in the low-silica region will eventually be monoxide (“FeO” precipitates). The fayalite (2FeO.SiO₂) primary phase field is also expected to be present at very low copper concentration. In the copper-free system, the fayalite is a congruently melting compound at 1478 K (1205 °C) and is bounded by binary eutectics at 1461 K and 1459 K (1188 °C and 1186 °C) on the iron- and silica-rich sides, respectively.^[12] The exact composition where the transition from spinel to monoxide occurs in the Cu-Fe-O-Si system, and the boundaries with the fayalite primary phase field, is yet to be determined.

The relative effects of Al, Ca, Mg, and Fe oxide additions on the 1573 K (1300 °C) liquidus isotherm in the tridymite primary phase field are illustrated in Figure 11. The substitution of copper oxide with the above metal oxides strongly stabilizes the high-silica liquid phase relative to the tridymite solid with the strongest effect introduced by the Al, followed by the Ca, Mg, and Fe oxides.

D. Composition of Tridymite Crystals in the Cu-Me-O-Si (Me = Ca, Mg, Al, and Fe) in Equilibrium with Metallic Copper

The compositions of tridymite crystals in equilibrium with metallic copper and liquid slag in the Cu-Me-O-Si system are summarized in Figure 12. Figures 12(a) and (b) indicate that the concentrations of CaO and MgO in solution in tridymite are negligible. Figures 12(c) and (d), on the other hand, show a significant dissolution of

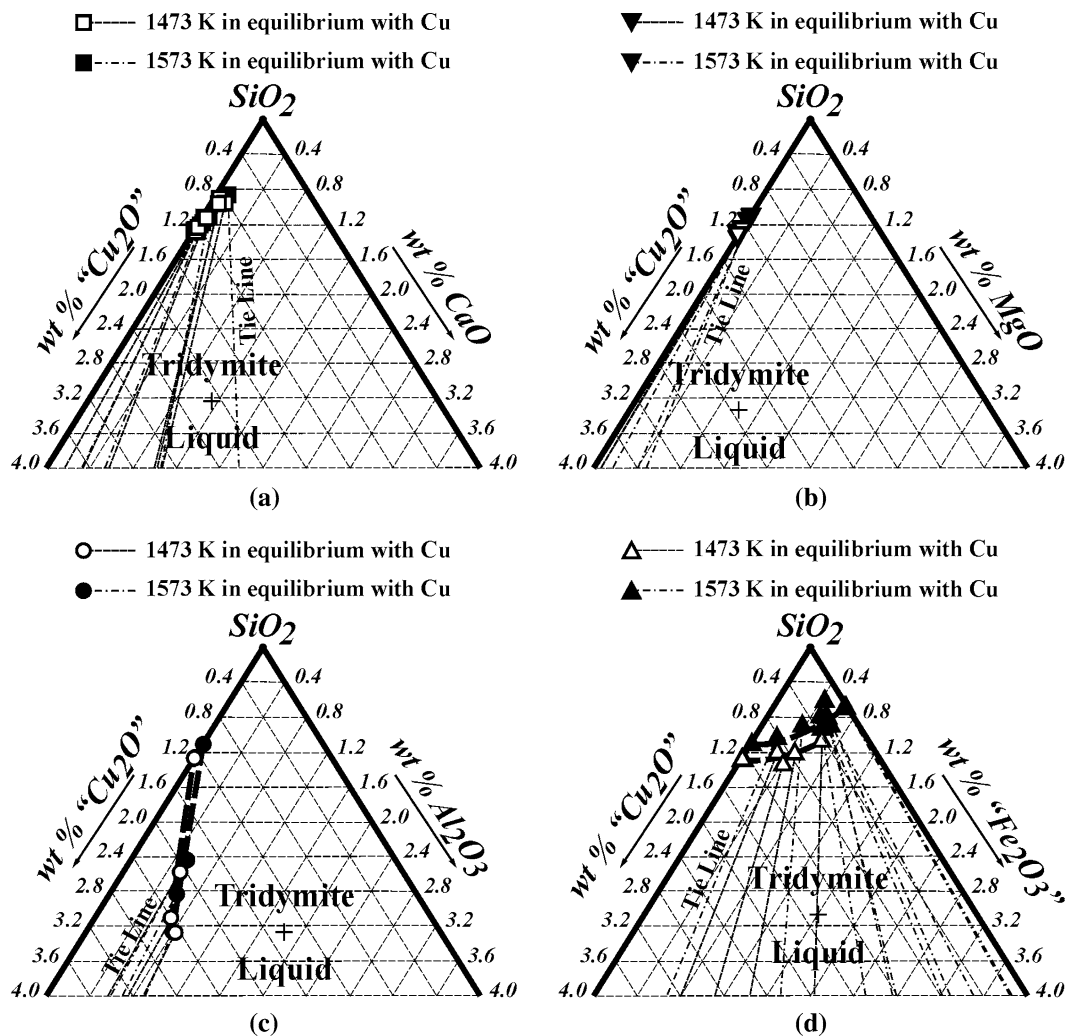


Fig. 12—Tridymite compositions and tie lines to the corresponding liquidus compositions in equilibrium with metallic copper between 1473 K and 1573 K (1200 °C and 1300 °C) in (a) “ Cu_2O ”- SiO_2 - CaO system, (b) “ Cu_2O ”- SiO_2 - MgO system, (c) “ Cu_2O ”- SiO_2 - Al_2O_3 system, and (d) “ Cu_2O ”- SiO_2 -“ Fe_2O_3 ” system.

Al_2O_3 and “ Fe_2O_3 ” in the tridymite crystals; the relationship between copper solubility and the third component is different in each of the cases. An increase in the Al_2O_3 concentration in the tridymite crystal is accompanied by an increase in “ Cu_2O ” concentration, while an increase in the “ Fe_2O_3 ” concentration in the tridymite is associated with a decrease in “ Cu_2O ” concentration.

IV. CONCLUSIONS

The phase equilibria and liquidus temperatures in the Cu-O-Si systems have been determined at two different conditions, *i.e.*, in air and in equilibrium with metallic copper. The compositions of the liquid and solid phases in the tridymite primary phase field and the eutectic temperatures of the systems at the two conditions have been measured using electron probe X-ray microanalysis

(EPMA) techniques. The effects of CaO , MgO , Al_2O_3 , and “ Fe_2O_3 ” on the compositions of liquid and tridymite phases of the system in equilibrium with metallic copper have been investigated at 1473 K and 1573 K (1200 °C and 1300 °C).

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